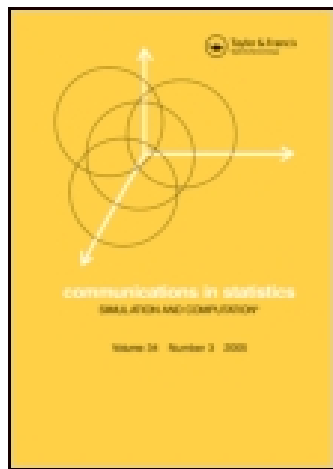


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Using the bootstrap in testing symmetry versus asymmetry

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USING THE BOOTSTRAP IN TESTING
SYMMETRY VERSUS ASYMMETRY

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Key Words and Phrases: empirical cdf, nonparametric test, testing symmetry, center of symmetry, symmetric bootstrap, estimating p-values.

ABSTRACT

One approach to testing the hypothesis of symmetry versus asymmetry is to use estimates of the center of symmetry in nonparametric tests about a known center. The problem with this approach is that the resulting tests are not distribution-free. Hence, one cannot compute critical values, p-values, or probabilities of errors for these tests. We propose to sidestep this problem with a symmetric bootstrap procedure which uses bootstrap samples from (a smoothed version of) the closest symmetric distribution to the empirical distribution of the data to estimate these values. In this paper, we report simulation experience with this approach in testing the hypothesis of symmetry using the Schuster and Narvarte estimator of the center of symmetry in the so-called Butler sup norm test for symmetry about a known center.

1. INTRODUCTION

Let $X_1, \dots, X_n$ be the order statistics of a random sample from a population having cumulative distribution function (cdf) F . In this paper, we consider the problem of testing the nonparametric hypothesis of symmetry versus asymmetry, i.e. we test

$$H_0: F \text{ is symmetric about some center } \theta \quad (1.1)$$

against

$$H_a: F \text{ is not symmetric,} \quad (1.2)$$

where both θ and F are unknown. Important reasons for general interest in testing this hypothesis can be found in Antille, et al. (1982).

One approach to tests of this hypothesis is to estimate θ by $\hat{\theta}^*$ and then use $\hat{\theta}^*$ in nonparametric tests for symmetry about a known θ , e.g. see Schuster and Narvarte (1973), Antille, et al. (1982), Boos (1982), and Koziol (1983). The problem with this approach is that the resulting tests are not distribution-free. Hence, one cannot compute critical values, p-values or probabilities of errors for these tests. We sidestep this problem by using a modified bootstrap procedure to estimate the associated critical and/or p-values.

Typically, the bootstrap is used to estimate the distribution of some functional, say $H = H(F, X_1, \dots, X_n)$. The bootstrap algorithm for N replications is then:

For $b = 1$ to N

- Sample $X_1^b, \dots, X_n^b$ (with replacement) from the empirical cdf F_n based on $X_1, \dots, X_n$.
- Compute $H_b = H(F_n, X_1^b, \dots, X_n^b)$.

Next b .

The usual bootstrap estimate of the distribution of $H = H(F, X_1, \dots, X_n)$ is the empirical distribution of $H_1, \dots, H_N$, with N large. A smooth version of F_n such as a kernel estimator of F (defined in (2.7)), is often used in the bootstrap algorithm when F is continuous (see Efron (1981), (1982)).

Suppose H_0 is true and we want to estimate the distribution of $H = H(F, X_1, \dots, X_n)$. Since F_n is not symmetric, bootstrap samples should not be taken from F_n , but rather from a symmetric distribution which is close to F_n . Efron (1979) suggests sampling from the symmetric cdf obtained from F_n by reflection about the median. However, one can easily see that this would not be appropriate when sampling from a distribution in which the median is not unique, e.g. when $F'(x)$ is zero in a neighborhood of the center of symmetry. In this case, the sample median would not converge to the center of symmetry in probability and the symmetric bootstrap cdf of (1.3), with θ^* equal to the sample median, would have too large a variance. Instead, we study bootstrap samples taken from the closest symmetric distribution to F_n in sup norm which Schuster (1986) has identified as the symmetric cdf $G_n = G_n(\cdot; \theta^*)$ obtained from F_n by reflecting $X_1, \dots, X_n$ about θ^* where θ^* (formally defined in 2.5) is the SN estimator (Schuster and Narvarte (1973)) of the center of symmetry. This procedure is what we call the **symmetric bootstrap**. Here, G_n is the empirical cdf of the $2n$ data points $X_1, \dots, X_n, 2\theta^* - X_1, \dots, 2\theta^* - X_n$. Notice that G_n is a symmetric distribution with center of symmetry θ^* and that we can write

$$\begin{aligned} G_n(x) &= \{F_n(x) + F_n^*(x)\}/2 \\ &= \{F_n(x) + 1 - F_n(2\theta^* - x)\}/2 \end{aligned} \quad (1.3)$$

where F_n and F_n^* are the empirical cdf's of the samples $X_1, \dots, X_n$ and $2\theta^* - X_1, \dots, 2\theta^* - X_n$, respectively. In the continuous case, we take our bootstrap samples from the smoothed version of G_n given by the kernel estimate in (2.6). Our motivation for this choice is discussed in Section 2 where we give the details of our bootstrap test for symmetry. In Section 3, we present some simulation results.

2. OUR BOOTSTRAP TEST FOR SYMMETRY

Two well-known tests for symmetry about a known θ are based on the so-called Butler (1969) statistic (see also Orlov (1972) and Smirnov (1947))

$$h_n(\theta) = \sup |F_n(x) - 1 + F_n((2\theta - x)-)|, \quad (2.1)$$

and the Rothman-Woodroffe (1972) statistic

$$h_n(\theta) = \int_{-\infty}^{\infty} (F_n(x) - 1 + F_n((2\theta - x)-))^2 dx. \quad (2.2)$$

As mentioned in Section 1, one natural approach to testing the nonparametric hypothesis of symmetry versus asymmetry, when θ and F are unknown, is to substitute a robust estimator θ^* in (2.1) or (2.2) and then use the statistic $h_n(\theta^*)$ to test the general hypothesis of symmetry. In either case, the resulting test statistic $h_n(\theta^*)$ is not distribution-free.

A second approach to testing the general hypothesis of symmetry is to use a test statistic which measures the distance in some norm from F_n to the closest symmetric distribution. If the distance measure is sup norm or integrated square error, and θ^* is the natural estimator of θ obtained by minimizing (2.1) or (2.2) over all θ , then the two approaches discussed are identical in the corresponding cases (see Schuster (1986)); that is, $G_n(x; \theta^*)$ of (1.3), with θ^* as in (2.5) ((2.3)), is the closest symmetric distribution to the empirical distribution function in sup norm (integrated square error).

Boos (1982) indicates that Knusel (1969) first showed that (2.2) is minimized over all θ by the HL (Hodges-Lehmann) estimator

$$\theta^* = \text{median} \{(X_i + X_j)/2, 1 \leq i, j \leq n\}. \quad (2.3)$$

Schuster and Narvarte (1973) have shown that (2.1) is minimized over all θ by any $\theta^* \in [m(L), M(L)]$ where for $k = 0, 1, \dots, n-1$ ($[x]$ is the greatest integer $\leq x$),

$$m(k) = \max \{(X_i + X_j)/2: 1 \leq i \leq [(n-k+1)/2], j=n-k+1-i\},$$

$$M(k) = \min \{(X_i + X_j)/2: k+1 \leq i \leq [(n+k+1)/2], j=n+k+1-i\},$$

and

$$L = \min \{k: m(k) \leq M(k)\}. \quad (2.4)$$

The natural unbiased estimator of θ in this case is the SN estimator

$$\theta^* = [m(L) + M(L)]/2. \quad (2.5)$$

The asymptotic distribution of θ^* is non-normal and is given in Rao, Schuster and Littell (1975). However, the SN estimator is robust in the sense that $n^{1/2}(\theta^* - \theta)$ converges to zero for any

$\delta > 0$ if F is continuous and symmetric about θ . Schuster and Narvarte show that if F is symmetric and n large, then at least 90% of the time, $L \leq 2n^{1/2}$. In addition, Theorem 1 of Schuster and Narvarte indicates that the following are equivalent (L as in (2.4)):

- (i) a minimizes (2.1),
- (ii) $m(L) \leq a \leq M(L)$,
- (iii) $nh_n(a) = L$.

Thus the value of $L = nh_n(\theta^*)$ can easily be obtained from the procedure above while computing the estimator θ^* . On the surface it appears that the estimator θ^* of (2.5) and $h_n(\theta^*)$ of (2.1) can be computed faster, using less storage, than θ^* of (2.3) and $h_n(\theta^*)$ of (2.2). For these reasons, particularly the computational advantages, as well as our own interest in the SN estimator, we have chosen to study the performance of the test statistic $nh_n(\theta)$ of (2.1) using $\theta = \theta^*$ of (2.5) in our bootstrap test. (Here and in the remaining parts of this paper θ^* will refer to the SN estimator θ^* of θ given in (2.5).)

We want a test which rejects the null hypothesis of symmetry if and only if

$$L = L(F_n, X_1, \dots, X_n) = nh_n(\theta^*) \geq l$$

where

$$P_F(L \geq l; H_0 \text{ is true}) = \alpha(l; F)$$

is the probability of an error of type I for the test at l and F , for $l = 0, 1, 2, \dots, n$. Since $\alpha(l) = \alpha(l; F)$ will depend on F , we use our symmetric bootstrap to approximate the distribution of L and $\alpha(l)$ under H_0 as follows:

For $b = 1$ to N , let $L_b = L(G_n, X_1^b, \dots, X_n^b)$ be the value of L obtained from the b^{th} bootstrap sample of size n from G_n . Then the empirical cdf of $L_1, \dots, L_N$ estimates the distribution of L and for $l = 0, 1, 2, \dots, n$,

$$\begin{aligned} \alpha_B(l) &= \# \{L_b \geq l\} / N \\ &= \text{the proportion of } L_b \text{'s} \geq l, \end{aligned}$$

estimates $\alpha(\mathcal{L})$. If $\tilde{\mathcal{L}} = L(F_n, X_1, \dots, X_n)$ is the value of L obtained from the original sample, then $\alpha_B(\tilde{\mathcal{L}})$ is the approximate p-value for the test.

While running our simulation experiments, we suspected that improved estimates of p-values would be obtained when sampling from continuous distributions if our bootstrap samples came from a nearby continuous distribution instead of discrete G_n . This indeed turned out to be the case and so we replaced G_n by the following nearby continuous symmetric kernel cdf G_n^* in all our simulations reported in Section 3:

$$G_n^*(x) = \sum K((x - Y_i)/a)/2n \quad (2.6)$$

where the summation is over $Y_1, \dots, Y_{2n}$ consisting of the $2n$ data points $X_1, \dots, X_n$ and $2\theta^* - X_1, \dots, 2\theta^* - X_n$, K is the cdf of the uniform over $[-1, 1]$, and positive a is the smoothing parameter. Note that

$$G_n^*(x) = \{\hat{F}_n(x) + 1 - \hat{F}_n((2\theta^* - x)-)\}/2$$

where

$$\hat{F}_n(x) = \sum K((x - X_i)/a)/n \quad (2.7)$$

is the usual Parzen-Rosenblatt kernel estimator of F .

Based on our experiments and theoretical considerations (see, Schuster (1975)), we feel that a good choice of the smoothing parameter would be a value which makes the derivative of (2.7) a good estimate of the derivative of F . The method of cross-validation, for example, works well for choosing the smoothing parameter a for short-tailed distributions, like the normal. For the uniform kernel, one should think of $2a$ as the common length of the class interval in constructing a regular histogram of the data. For more detailed guidelines for selecting the smoothing parameter a , see Devroye, et al. (1985). However, results did not turn out to be very sensitive to reasonable choices of the value of a , at least for the symmetric distributions studied. In order to simplify matters, we used prior experience with density estimators in choosing $a = 0.5$ in all the simulations reported in Tables 1-4. In Section 3, we discuss our method of choosing the smoothing parameter a for the Chi-square distributions of Table 5.

Remark. Note that sampling from the continuous cdf in (2.6) is easy since G_n^* can be expressed as the convolution of G_n with the cdf of a uniform over $[-a,a]$. Thus one can sample Z from (2.6) by $Z = Y + aU$ where Y and U are independent, Y has cdf G_n , and U is uniform over $[-1,1]$.

We report our simulation results in the next section.

TABLE 1
SYMMETRIC DISTRIBUTIONS
($n = 25$, $N = 10,000$)

	NORMAL		DBLE. EXPO.		CAUCHY	
ℓ	$\alpha_B(\ell)$	$\alpha_S(\ell)$	$\alpha_B(\ell)$	$\alpha_S(\ell)$	$\alpha_B(\ell)$	$\alpha_S(\ell)$
1	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
2	.9885	.9892	.9894	.9907	.9902	.9914
3	.6454	.6490	.6424	.6485	.6722	.7026
4	.1846	.1849	.1707	.1851	.2033	.2632
5	.0411	.0405	.0361	.0418	.0419	.0713
6	.0032	.0033	.0025	.0045	.0032	.0110
7	.0002	.0002	.0002	.0003	.0001	.0014
8	.0000	.0000	.0000	.0000	.0000	.0001
L	L_B	L_S	L_B	L_S	L_B	L_S
AVE L	2.863	2.867	2.841	2.871	2.911	3.041
STD L	.8340	.8307	.8106	.8354	.8336	.9297
	$\tilde{\ell} = 3$		$\tilde{\ell} = 2$		$\tilde{\ell} = 4$	
	p-value estimates					
	$\alpha_S(3) = .6490$		$\alpha_S(2) = .9907$		$\alpha_S(4) = .2632$	
	$\alpha_B(3) = .6454$		$\alpha_B(2) = .9894$		$\alpha_B(4) = .2033$	

TABLE 2
SYMMETRIC DISTRIBUTIONS
 (n = 50, N = 10,000)

	NORMAL		DBLE. EXPO.		CAUCHY	
ℓ	$\alpha_B(\ell)$	$\alpha_S(\ell)$	$\alpha_B(\ell)$	$\alpha_S(\ell)$	$\alpha_B(\ell)$	$\alpha_S(\ell)$
2	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
3	.9714	.9691	.9718	.9738	.9745	.9800
4	.7181	.7243	.7254	.7368	.7632	.7657
5	.3479	.3565	.3480	.3668	.4231	.4330
6	.1384	.1501	.1392	.1558	.2013	.2061
7	.0389	.0415	.0383	.0446	.0737	.0798
8	.0099	.0113	.0095	.0140	.0244	.0285
9	.0018	.0021	.0017	.0022	.0059	.0066
10	.0006	.0004	.0003	.0005	.0012	.0010
11	.0001	.0000	.0002	.0000	.0002	.0001
12	.0000	.0000	.0001	.0000	.0001	.0001
13	.0000	.0000	.0000	.0000	.0000	.0001
L	L_B	L_S	L_B	L_S	L_B	L_S
AVE L	4.227	4.255	4.235	4.295	4.468	4.501
STD L	1.169	1.192	1.162	1.198	1.316	1.330
	$\tilde{\ell} = 5$		$\tilde{\ell} = 3$		$\tilde{\ell} = 4$	
	p-value estimates					
	$\alpha_S(5) = .3565$		$\alpha_S(3) = .9738$		$\alpha_S(4) = .7657$	
	$\alpha_B(5) = .3479$		$\alpha_B(3) = .9718$		$\alpha_B(4) = .7632$	

TABLE 3
SYMMETRIC DISTRIBUTIONS

($n = 100$, $N = 10,000$)

	NORMAL		DBLE. EXPO.		CAUCHY	
l	$\alpha_B(l)$	$\alpha_s(l)$	$\alpha_B(l)$	$\alpha_s(l)$	$\alpha_B(l)$	$\alpha_s(l)$
3	1.0000	1.0000	1.0000	.9999	1.0000	1.0000
4	.9858	.9873	.9860	.9862	.9900	.9874
5	.8673	.8603	.8627	.8722	.8887	.8873
6	.6324	.6316	.6284	.6455	.6859	.6863
7	.3841	.3853	.3766	.3948	.4492	.4555
8	.2001	.2022	.2008	.2207	.2782	.2813
9	.0911	.0932	.0937	.1101	.1480	.1528
10	.0396	.0422	.0397	.0501	.0746	.0837
11	.0147	.0162	.0159	.0181	.0329	.0383
12	.0046	.0054	.0057	.0072	.0146	.0164
13	.0016	.0015	.0021	.0022	.0056	.0067
14	.0004	.0007	.0006	.0008	.0020	.0028
15	.0000	.0004	.0000	.0001	.0008	.0008
16	.0000	.0000	.0000	.0000	.0002	.0003
17	.0000	.0000	.0000	.0000	.0001	.0000
L	L_B	L_s	L_B	L_s	L_B	L_s
AVE L	6.222	6.226	6.212	6.308	6.571	6.600
STD L	1.648	1.672	1.666	1.719	1.870	1.918
	$\tilde{l} = 4$		$\tilde{l} = 6$		$\tilde{l} = 9$	
	p-value estimates					
	$\alpha_s(4) = .9873$		$\alpha_s(6) = .6455$		$\alpha_s(9) = .1528$	
	$\alpha_B(4) = .9858$		$\alpha_B(6) = .6284$		$\alpha_B(9) = .1480$	

TABLE 4
ASYMMETRIC DISTRIBUTIONS
 (n = 100, N = 10,000)

l	LOGNORMAL		EXPONENTIAL	
	$\alpha_B(l)$	$\beta_S(l)$	$\alpha_B(l)$	$\beta_S(l)$
3	0.9999	1.0000	1.0000	1.0000
4	0.9861	1.0000	0.9881	1.0000
5	0.8734	1.0000	0.8800	1.0000
6	0.6557	1.0000	0.6683	1.0000
7	0.3992	1.0000	0.4172	1.0000
8	0.2242	1.0000	0.2505	1.0000
9	0.1058	1.0000	0.1312	0.9999
10	0.0470	1.0000	0.0657	0.9997
11	0.0195	0.9998	0.0261	0.9991
12	0.0074	0.9994	0.0117	0.9968
13	0.0025	0.9973	0.0040	0.9871
14	0.0010	0.9920	0.0016	0.9623
15	0.0005	0.9761	0.0005	0.9115
16	0.0001	0.9370	0.0002	0.8286
17	0.0000	0.8666	0.0001	0.7000
18	0.0000	0.7592	0.0001	0.5554
L	L_B	L_S	L_B	L_S
AVE L	6.3223	19.0683	6.4453	17.7986
STD L	2.6387	2.3186	2.6001	2.4335
		$\tilde{l} = 20$	$\tilde{l} = 18$	
		p-value ≈ 0.0000	p-value ≈ 0.0001	

TABLE 5
CHI-SQUARE DISTRIBUTIONS

(n = 100, N = 10,000)

df	θ	a	$\bar{L}_s$	$S(L_s)$	l	$\alpha_B(l)$	$\alpha_s(l;H)$	$\beta_s(l)$
4	3.7447	1.0	13.3	2.5	9	.1040	.1216	.9774
					10	.0492	.0626	.9421
					11	.0195	.0260	.8721
					12	.0080	.0127	.7658
6	5.7313	1.5	11.3	2.4	13	.0027	.0046	.6250
					9	.1066	.1109	.8769
					10	.0513	.0527	.7678
					11	.0199	.0208	.6153
8	7.7244	2.2	10.2	2.4	12	.0082	.0088	.4605
					13	.0024	.0037	.3090
					8	.2159	.2121	.8664
					9	.1038	.1027	.7416
10	9.7203	2.6	9.5	2.4	10	.0493	.0480	.5957
					11	.0184	.0210	.4296
					12	.0076	.0081	.2860
					7	.3916	.3847	.9028
12	11.7176	3.0	9.0	2.3	8	.2219	.2072	.7933
					9	.1064	.0997	.6426
					10	.0478	.0452	.4809
					11	.0179	.0170	.3204
20	19.7123	4.2	7.9	2.2	12	.0071	.0074	.1969
					7	.3790	.3744	.8616
					8	.1988	.2073	.7236
					9	.0871	.0932	.5520
30	29.7097	5.3	7.3	2.1	10	.0376	.0418	.3906
					11	.0149	.0156	.2497
					12	.0063	.0058	.1408
					7	.4035	.3721	.7150
					8	.2347	.2063	.5431
					9	.1180	.0963	.3661
					10	.0553	.0438	.2296
					11	.0256	.0165	.1241
					12	.0113	.0072	.0660
					7	.3731	.3775	.6140
					8	.2009	.2070	.4267
					9	.0911	.0893	.2667
					10	.0408	.0422	.1538
					11	.0153	.0144	.0753
					12	.0048	.0056	.0339

3. SIMULATION RESULTS

Simulation results are summarized in Tables 1-5. Simulations were run on an IBM 4341 (32 bit word) with the n^{th} uniform U_n defined by $U_n = I_n/P$ with $P = 2^{31} - 1$, and I_n generated by the multiplicative congruential generator $I_n = (I_{n-1} * 16807) \bmod P$. In Tables 1-3, we considered the statistician's prototypes of symmetric short (normal), medium (double exponential), and heavy tailed (Cauchy) distributions to find how well the test would estimate its p-values and probabilities of errors of type I. Most of our studies centered around samples of size $n = 100$ from these distributions with $N = 10,000$ replications. However, we have included Tables 1 and 2 for samples of sizes $n = 25$ and $n = 50$ to give the reader an idea of the amount of variability for different n . Note that 10,000 samples of n are used for our regular simulation estimate $\alpha_s(l)$ of the p-value of the test at l , while only one sample of n is used for the bootstrap estimate $\alpha_b(l)$, as would be the case in most applications. Comparing $\alpha_b(l)$ with $\alpha_s(l)$, the test seems to estimate its p-values and probabilities of error of type I exceptionally well. Results reported here are consistent with results obtained from replicating runs. Samples from a contaminated normal (95% $N(0,1)$, 5% $N(0,10)$) exhibited almost the same behavior as the samples from the standard normal and were omitted.

The numbers below L_b and L_s are the average values (AVE) and standard deviations (STD) of L for the 10,000 bootstrap and regular simulations, respectively. The values of $\tilde{l}$ are the observed values of L for the single underlying sample in each case. At the bottom of each table we give the estimated p-values for the samples from the three symmetric distributions. Notice that one would fail to reject the hypotheses of symmetry for all nine distribution samples in Tables 1-3 for levels below the 15% level. The average value and standard deviation of L are reasonably constant over each of these symmetric distributions.

Tables 4 and 5 report experience with asymmetric distributions. Table 4 summarizes simulations from the lognormal

and the exponential distributions. The results are typical of estimates of p-values (essentially zero) obtained from these highly asymmetric distributions. In Table 5, we use samples of $n = 100$ from the Chi-square distribution for select degrees of freedom (df) between four and thirty to study the performance of the test as one goes from a highly asymmetric distribution to a nearly symmetric distribution. For asymmetric distributions, we use $\hat{\alpha}_S(l)$ to report the estimate of the power of the test at l (which rejects H_0 if $L \geq l$), based on regular simulations with 10,000 replications in both Table 4 and Table 5.

For asymmetric F , the closest symmetric distribution to F in sup norm is

$$H(x; \theta) = [F(x) + 1 - F((2\theta - x)^-)]/2$$

where $\theta = \theta(F)$ minimizes

$$h(a) = \sup_x |F(x) - H(x; a)| \quad (3.1)$$

over all a (Schuster (1986)). For these asymmetric cases, θ^* estimates θ , and under mild regularity conditions (i.e., $f(x) = F'(x) > 0$ on $S = \{s: 0 < F(s) < 1\}$), $\sqrt{n}(\theta^* - \theta) = O_p(1)$. It then follows that for asymmetric F , $\alpha_B(l) = \alpha_B(l; F)$ is both a regular simulation estimate of the p-value at l of a continuous symmetric distribution (namely, $G_n^*(\cdot; \theta^*)$ of (2.6)), as well as a bootstrap estimate of $\alpha(l; H)$, the p-value at l of the closest symmetric distribution H to F . In the asymmetric case, we will call $\alpha(l) = \alpha(l; H)$ the actual (theoretical) p-value of the test.

For the Chi-square distribution experiments reported in Table 5, we demonstrate that $\alpha_B(l)$ approximates $\alpha_S(l; H)$ (and hence $\alpha(l; H)$). For each degree of freedom (df) reported, we give: the value of θ minimizing (3.1) (which θ^* is estimating) obtained by a numerical routine (described in Schuster (1986)); the value of the smoothing parameter a used in the bootstrap simulations; the average L , denoted $\bar{L}_S$; and the sample standard deviation of L_S , denoted $S(L_S)$, for $N = 10,000$ replications of the test statistic L . We have included an $N = 10,000$ regular simulation estimate $\alpha_S(l; H)$ of $\alpha(l; H)$ in order to demonstrate that $\alpha_B(l)$ estimates $\alpha(l; H)$. Here as in Tables 1-4, the bootstrap estimates $\alpha_B(l)$ are

based on a single sample of 100 (the first sample of 100 used for the 10,000 regular simulations of L used in estimating $\alpha(\ell; H)$ and $\beta(\ell)$) with 10,000 bootstrap replications.

Ideally, as mentioned in Section 2, a data-driven method should be used to choose the value of the smoothing parameter a of the bootstrap distribution G_n^* of (2.6) when testing the hypothesis of symmetry and sampling is from a continuous distribution. However, to simplify computations for Table 5, for each df we used the value of a (proportional to $n^{-1/5}$), given by the formulas in Parzen (1962) to minimize the mean square error of the (uniform) kernel estimate of the density for a sample of $n = 100$. The accuracy with which $\alpha_B(\ell)$ estimates the p -value, $\alpha(\ell)$, seems to be much more sensitive to the proper calibration of the smoothing parameter (according to the guidelines following (2.7)) in the asymmetric case than in the symmetric case.

4. SUMMARY

In this paper we establish the potential of testing the general hypothesis of symmetry using symmetric bootstrap procedures. The particular test investigated here (based on the use of the Schuster and Narvarte (SN) estimator θ^* of the center of symmetry θ in the Butler sup norm statistic $h_n(\theta)$ of (2.1)) seems to estimate its associated p -values and probabilities of errors of type I quite well. Surprisingly enough, errors in estimating p -values and probabilities of type I errors appeared closer to order $N^{-1/2}$ than to $n^{-1/2}$ where $N = 10,000$ is the number of symmetric bootstrap samples and n is the sample size. In fact, we see that $|\alpha_B(\tilde{\ell}) - \alpha_S(\tilde{\ell})| \leq N^{-1/2}$ for seven of the nine values of $\tilde{\ell}$ given at the bottom of Tables 1-3. The exceptions are: the sample of size 25 from the Cauchy distribution in Table 1 when $\tilde{\ell} = 4$ and $\alpha_S(4) = .2632$ and $\alpha_B(4) = .2033$; and the sample of size 100 from the double exponential distribution in Table 3 when $\tilde{\ell} = 6$ and $\alpha_S(6) = .6455$ and $\alpha_B(6) = .6284$. However, we used the constant value $a = 0.5$ for the smoothing parameter in all the

simulations reported (for samples of size $n = 25, 50, 100$) in Tables 1-3. As mentioned in Section 2 and 3, data driven methods should be used to calibrate the choice of smoothing parameter.

Although the test is not distribution-free, a glance at Tables 1-3 indicates that critical values and probabilities of errors of type I do not vary widely across these common, unimodal, symmetric distributions.

Finally, the choice of the SN estimator θ^* of θ in the Butler statistic $h_n(\theta)$ was based on our own interest in this estimator, the fact that θ^* minimizes $h_n(\theta)$ over all θ (Schuster and Narvarte (1973), and the fact that θ^* is the center of the closest (in sup norm) distribution function to the empirical distribution function (Schuster (1986)). A computationally simpler estimate of θ such as the sample median used in $h_n(\theta)$ might perform as well.

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